Mathematics 1
Sheet No.:1
Title: Derivatives

Textbook: Calculus, Early Transcendental Functions, Fourth Edition, by: Robert T. Smith, Roland B. Minton, McRraw-Hill Publications.

Find the first derivative of the following functions

(1) $f(x) = x^7 - 3\sqrt{x} + \sin(x^2+1) + \frac{1}{x^9} + 21$

(2) $f(x) = \sin^{-1}(x^3 + 4x - 1) + 3x^2 - 7$

(3) $f(x) = (x^3+1)\sec^{-1}(5x)$

(4) $f(x) = \frac{\tan x}{1+\sec x}$

(5) $f(x) = \frac{3e^{2x}+1}{\sin^{-1} x}$

(6) $f(x) = e^{\tan x} + 3x^2 + 4$

(7) $f(x) = \sin(\cos^{-1} x)$

(8) $f(x) = \frac{\cot x}{1+e^{x+3}}$

Find the first and the second derivatives of the following functions

(9) $f(x) = (\sin x + 1)^4$

(10) $f(x) = \cos(x^3+1)$

Find the first derivative y' of the following (with respect to x),

(11) $\tan x + \sec y + x^4 = y^2$

(12) $y^3 + e^{x^2+1} = 3e^y + 7$

(13) $x^3 y^5 + \sec^{-1} y + e^{\sin x} - 7 = 0$

(14) $y = x^2 \sin y$

(15) $x^2 + \tan^{-1} y = y^2 + 4x + 8$

By using the chain rule calculate $\frac{dy}{dx}$ for the following,

(16) $y = \tan^{-1}(t+1)$, $x = t^2 + 3t + 4$

(17) $y = 3t + 1$, $x = t^2 + 4$

(18) $y = (3t+1)$, $x = t^2 + 3t + 4$

(19) $y = u^2 + 2u + 1$, $x = 2u + 4$

Find the first derivative y' of following by using the logarithmic differentiation,

(20) $y = x^{\sin^{-1} x}$

(21) $y = (1 + x^2)^{\sec x}$

(22) $y = \frac{(x+1)^3 (x^2 + 8x - 3)^5}{(x-1)^7 (x^3 + 2)^4}$

(23) $y = \frac{(x^2 + 1)^4 \sinh x}{(x^5 + 3)^8}$